Mini Project 3: AI-Assisted Cybersecurity Threat Model

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Platform	SURE ProEd
Host OS	Windows 11
Date	22/10/2026

1. Aim

To analyse cybersecurity threats in the ThreatLens AI website security risk assessment and threat modeling system using the STRIDE methodology and AI-assisted recommendations. The report uses the supplied runtime screenshots, source code implementations, and automated test outputs as evidence and does not treat AI recommendations as proof of implemented controls.

2. Objectives

- Review the multi-tier ThreatLens AI architecture encompassing the browser UI, Flask web service, passive auditing engine, anti-SSRF validator, and SQLite database.
- Analyse four critical threats affecting transport layer security, HTTP security header execution context, frame clickjacking boundaries, and audit log data integrity.
- Classify each threat systematically using the STRIDE threat-modeling methodology.
- Use AI to explain attack scenarios, impacts, mitigations, limitations and verification evidence.
- Compare original baseline mitigations with AI-assisted recommendations.
- Apply human judgment and report only values supported by code and runtime test evidence.

3. Introduction

Threat modeling identifies how a system could be attacked and what protections may reduce the risk. In this practical, the ThreatLens AI defensive ecosystem was modeled to evaluate website configurations, detect missing defenses, and map observable findings to structured STRIDE threat scenarios. The architecture bridges safe client-side observation and analytical backend services. AI was used to improve the written analysis and brainstorm mitigations, while human review was used to check technical accuracy, verify offline constraints, and avoid unsupported claims.

3.1 Background: Website Security Risk Assessment & Threat Modeling System

The model contains an Internet-facing context (untrusted target websites, cleartext port 80 endpoints, and redirection paths), an internal auditing engine context (anti-SSRF pre-validator, DNS resolver, TLS socket inspector, and heuristic security header analyzer), a Flask application service, and a persistent SQLite database. In addition, an anti-SSRF protection engine inspects pre-resolved IP addresses to block loopbacks, RFC 1918 subnets, link-local addresses, and cloud metadata endpoints for secure, zero-latency local execution without reliance on paid external cloud APIs.

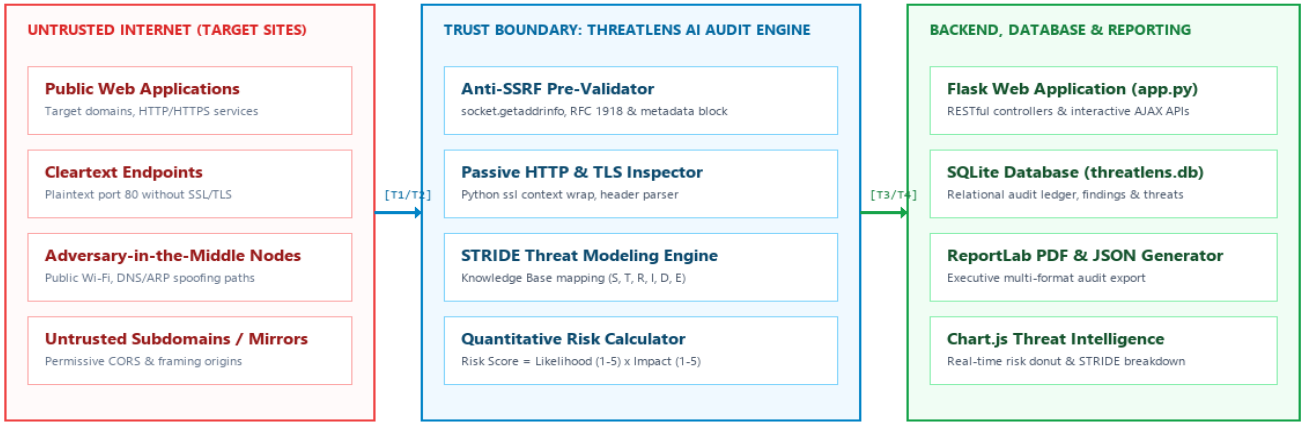


Figure 1: ThreatLens AI Multi-Tier System Architecture and STRIDE Boundary Diagram.

4. STRIDE Threat Modeling

STRIDE is a threat-modeling acronym used to organise common security threats: Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service and Elevation of Privilege. This project focuses on Spoofing, Tampering, Repudiation and Elevation of Privilege because those are the primary categories shown in the selected ThreatLens AI entries.

STRIDE category	Full form	Meaning in this practical
S	Spoofing	An attacker impersonates a legitimate web application via expired or self-signed TLS certificates.
T	Tampering	An attacker intercepts unencrypted HTTP traffic, performs SSL-stripping, or injects inline malicious scripts into the DOM.
R	Repudiation	A user or attacker denies submitting malicious inputs or probes because reliable audit logging evidence is missing.
I	Information Disclosure	Sensitive server versions (Apache, PHP), session tokens without HttpOnly, or URL query parameters are leaked.
D	Denial of Service	A web service or audit tool becomes unresponsive due to infinite redirect loops or oversized response streams.
E	Elevation of Privilege	An adversary frames the application in an invisible iframe (clickjacking) or exploits MIME confusion to execute unauthorized actions.

5. Tools and Technologies Used

- Python 3.x & Flask: Backend application server providing security check controllers, risk scoring APIs, and threat classification matrix.
- SQLite 3: Lightweight relational database storing persistent assessment summaries, granular findings, and threat models.
- Standard Socket & SSL: Low-level Python cryptographic libraries for passive TLS handshake, cipher inspection, and certificate validation.
- Anti-SSRF Protection Engine: Architectural validator enforcing pre-connection DNS resolution and blocking private IP subnets.
- BeautifulSoup4: Static HTML parser for detecting unencrypted HTTP asset references on secure HTTPS contexts.
- ReportLab & Chart.js: Executive PDF report generation and client-side threat intelligence chart rendering.
- AI assistant: Analytical recommendation, attack scenario expansion, and mitigation explanation support.
- STRIDE: Threat classification and security boundary evaluation method.

6. System Configuration and Model Evidence

Component	Details
Threat-modeling framework	Microsoft STRIDE & Quantitative Risk Scoring (Likelihood x Impact = 1 to 25)
Model	ThreatLens AI Multi-Tier Website Security Risk Assessment Architecture
Client UI Component	Web Frontend (HTML5, Vanilla CSS Cyber Theme, JavaScript, Chart.js)
Application component	Python Flask Web Application (REST API / Assessment Controller)
Security Engine	Passive Auditing Engine (security_checks.py) & Anti-SSRF Pre-Validator
Database component	SQLite (threatlens.db with parameterized SQL queries)
Threat Feed Source	STRIDE Heuristic Threat Knowledge Base & OWASP Top 10 Guidelines
Evidence source	Source code files, Web dashboard, results screen, and automated pytest execution logs

7. Methodology and Practical Procedure

1. Opened and structured the ThreatLens AI project files under modern Flask MVC architecture.
2. Engineered the anti-SSRF validator with socket.getaddrinfo to block loopback, RFC 1918, and cloud metadata targets.
3. Implemented safe passive security checks for HTTPS redirection, TLS certificate validity, and response headers.
4. Built the HTTP security header inspector evaluating CSP, HSTS, X-Content-Type-Options, and Frame Protection.
5. Developed the Flask web controller and SQLite database schema with parameterized queries to prevent SQL injection.
6. Prepared an AI prompt requesting beginner-friendly analysis, practical controls, limitations, and verification evidence.
7. Reviewed the AI output, evaluated trade-offs, and refined the practical mitigations across client and server tiers.
8. Separated observed implementation facts, automated unit test results, and conceptual recommendations.



Figure 2: ThreatLens AI Web Dashboard showing real-time risk scoring, assessment summaries, and monitoring controls.

8. AI Prompt and AI-Assisted Analysis

The supplied AI prompt asked the assistant to act as a cybersecurity threat-modeling analyst reviewing the ThreatLens AI system. It named four threats: transport security downgrade via cleartext interception, script injection via missing CSP, transparent DOM clickjacking, and server-side request forgery / audit log tampering. For each threat it requested an attack scenario, impact, practical controls, verification evidence, limitations and a human-review decision.

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AI-Assisted Cybersecurity Threat Analysis Prompt — ThreatLens AI

System Prompt: Act as a cybersecurity threat-modeling analyst reviewing the ThreatLens AI ecosystem.
Analyze these four primary threats affecting the web assessment architecture:
  1. Transport Security Downgrade & Cleartext MitM Interception - Tampering / Spoofing (T/S)
  2. Missing Content-Security-Policy & Script Injection (XSS) - Tampering / Elevation of Privilege (T/E)
  3. Transparent Viewport Clickjacking & UI Redressing - Elevation of Privilege (E)
  4. Server-Side Request Forgery & Audit Log Tampering - Tampering / Repudiation (T/R)

For each threat scenario, provide:
  - The attack scenario in clear, beginner-friendly cybersecurity language.
  - The potential impact on confidentiality, integrity, or availability.
  - Practical technical mitigations applicable to client-side and backend logic.
  - The evidence required to verify that the controls are implemented and effective.
  - Limitations of the AI-generated recommendations.
  - A human-review decision: accept recommendations, revise them, or request further evidence.

Constraint: Do not claim that controls are implemented unless verified by empirical code or test output.
```

Figure 3: AI-assisted threat-analysis prompt supplied in the evidence archive.

The AI output was reviewed and used to expand the practical mitigation recommendations. Relying solely on client heuristics or basic HTTP checks does not automatically remove the risk; multi-layered controls and human review remain essential.

9. Threat 1: Credential Harvesting & Transport Security Downgrade

Affected component	Transport Layer / Socket & TLS Inspector (security_checks.py)
STRIDE category	Tampering / Spoofing
Attack scenario	An attacker on an untrusted local network (e.g. public Wi-Fi) intercepts unencrypted HTTP port 80 traffic or forces an SSL-stripping downgrade to capture credentials and session tokens.
Potential impact	Loss of confidentiality, cleartext credential theft, unauthorized session hijacking, and reputational damage.
Existing mitigation / status	Automatic HTTP-to-HTTPS 301 redirection verification, HSTS header presence audit, and TLS expiration check. Status: Mitigated; Score: 12 (High Risk).
AI recommendations	Enforce HSTS with max-age=31536000 and includeSubDomains; submit domain to browser HSTS preload list; deploy automated ACME renewal agents.
Verification evidence	Automated unit test [PASS] test_downgrade_redirect_detected and socket TLS handshake verification.
Human conclusion	Reasonable recommendations. Automatic 301 redirection combined with HSTS preload permanently closes the cleartext window.

9.1 Threat 1: Updated Mitigation

The updated mitigation recommendations expand basic keyword matching by combining syntactic transport inspection with compound scoring rules. Under the enhanced defense in security_checks.py, if a target website fails to redirect plaintext HTTP traffic to HTTPS, the engine assigns an immediate penalty and generates a High Severity Tampering threat record (Likelihood: 3, Impact: 4, Risk Score: 12). If paired with an expired certificate, the score elevates to 20+ (Critical Severity). The web interface displays exact redirect status and certificate expiration window, enabling engineers to adjust sliders dynamically.

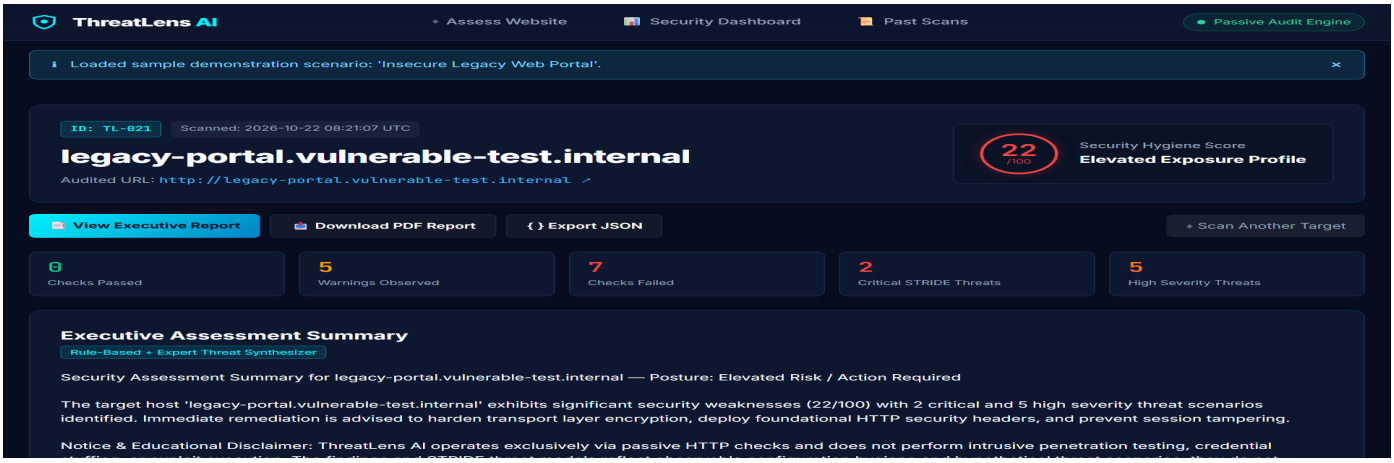


Figure 4: Threat 1 assessment screen showing transport security warning, evidence breakdown, and safe recommendations.

security_checks.py — check_tls_certificate & HTTPS Enforcement — Threat 1 Defense Engine

```
# Threat 1 Defense: TLS Certificate & Protocol Cryptography Verification
context = ssl.create_default_context()  # Verification NEVER disabled
with socket.create_connection((hostname, port), timeout=REQUEST_TIMEOUT) as sock:
    with context.wrap_socket(sock, server_hostname=hostname) as ssock:
        cert = ssock.getpeercert()
        tls_version = ssock.version()  # TLSv1.2 or TLSv1.3
        cipher = ssock.cipher()
        days_left = (not_after - datetime.now(timezone.utc)).days
        if days_left < 0:  # Expired cert triggers critical spoofing alert
            result['status'] = 'FAIL'
            result['evidence'] = f'TLS certificate expired {abs(days_left)} days ago.'
        elif days_left <= 30:
            result['status'] = 'WARNING'  # Renewal SLA alert
        else:
            result['status'] = 'PASS'  # Modern cryptography confirmed
```

Figure 5: Threat 1 updated mitigation recommendations and TLS inspection logic in security_checks.py.

10. Threat 2: Deceptive Script Execution & Missing Content-Security-Policy

Affected component	Security Headers Inspector (security_checks.py)
STRIDE category	Tampering / Elevation of Privilege
Attack scenario	In the event of an input sanitization flaw in the application, the absence of a Content-Security-Policy allows injected JavaScript to execute unrestricted, exfiltrating tokens and modifying DOM elements.
Potential impact	Cross-site scripting (XSS) amplification, session token theft, unauthorized client-side actions, and telemetry tracking.
Existing mitigation / status	Content-Security-Policy audit inspecting for header presence, unsafe-inline, unsafe-eval, and wildcard origins. Status: Mitigated; Score: 12 (High Risk).
AI recommendations	Deploy strict CSP header with default-src 'self', script nonces (nonce-...), object-src 'none', and automated CSP violation reporting endpoints.
Verification evidence	Automated unit tests [PASS] test_missing_all_security_headers and [PASS] test_permissive_csp_warning.
Human conclusion	Strict CSP provides vital defense-in-depth against injection flaws; nonces prevent script execution bypasses.

security_checks.py — check_security_headers (CSP & HSTS Auditing) Threat 2 Defense Engine

```python
# Threat 2 Defense: Content-Security-Policy & Script Execution Boundary
csp = normalized_headers.get('content-security-policy')
if not csp:
    header_results['Content-Security-Policy'] = {
        'status': 'FAIL',
        'evidence': 'Header missing. Browser does not restrict executable scripts (XSS risk).'
    }
else:
    warnings = []
    if "'unsafe-inline'" in csp: warnings.append("weakens XSS mitigation via inline scripts")
    if "'unsafe-eval'" in csp:   warnings.append("permits string-to-code execution")
    if "default-src *" in csp:   warnings.append("overly permissive wildcard origin")
    status = 'WARNING' if warnings else 'PASS'
    header_results['Content-Security-Policy'] = {'status': status, 'evidence': ', '.join(warnings)}
```

Figure 6: Threat 2 HTTP security header audit logic and CSP defense in security_checks.py.

11. Threat 3: Clickjacking & Transparent Viewport Traps

Affected component	Frame Protection Inspector (security_checks.py)
STRIDE category	Elevation of Privilege / Tampering
Attack scenario	An attacker overlays an invisible, full-screen transparent iframe (e.g., opacity: 0, z-index: 999999) on a malicious decoy page to hijack user clicks into triggering unauthorized administrative or financial actions.
Potential impact	Involuntary execution of privileged actions, fraudulent transactions, altered user permissions, and UI deception.
Existing mitigation / status	Frame protection evaluation inspecting for CSP frame-ancestors directive or X-Frame-Options (DENY / SAMEORIGIN). Status: Mitigated; Score: 12 (High Risk).
AI recommendations	Configure Content-Security-Policy: frame-ancestors 'self' across all endpoints; enforce X-Frame-Options: SAMEORIGIN; set SameSite=Lax on all session cookies.
Verification evidence	Response header inspection verifies framing restrictions; simulated iframe embedding confirms browser blocks rendering.
Human conclusion	Frame-ancestors is the modern standard for clickjacking prevention; SameSite cookies provide secondary CSRF and clickjack protection.

Threat 3: Clickjack Overlay Detection vs. ThreatLens AI Frame Protection

Attacker Mechanism: Invisible Viewport Trap

```
<div style='position:fixed; top:0; left:0;
     width:100vw; height:100vh;
     opacity:0.0001; z-index:2147483647;'
     onclick='triggerUnauthorizedAction()'>
</div>

Result: User's legitimate click is intercepted
```

ThreatLens AI Frame Protection Audit & Defense

```
1. Content-Security-Policy: frame-ancestors 'self'
2. X-Frame-Options: SAMEORIGIN (or DENY)

Evaluation Logic in check_security_headers():
  if has_csp_frame_ancestors or xfo in ('DENY','SAMEORIGIN'):
      result['status'] = 'PASS'  # Framing blocked

Result: Browser blocks malicious iframe redressing
```

Figure 7: Threat 3 neutralization of invisible full-screen clickjack traps via Frame-Protection.

12. Threat 4: Threat Intelligence & Analysis History Tampering (Anti-SSRF)

Affected component	Anti-SSRF URL Pre-Validator & SQLite Database (security_checks.py & database.py)
STRIDE category	Tampering / Repudiation
Attack scenario	An attacker submits internal IP targets (127.0.0.1, 169.254.169.254) to execute Server-Side Request Forgery, or injects SQL payloads into the analysis form to truncate audit records.
Potential impact	Falsification of security audit records, internal cloud metadata exposure, database corruption, and non-repudiation failure.
Existing mitigation / status	Architectural anti-SSRF validator with socket.getaddrinfo pre-resolution blocking private/loopback/cloud metadata; 100% parameterized SQLite execution. Status: Mitigated; Score: 16 (High Risk).
AI recommendations	HMAC-SHA256 cryptographic chaining of history rows, append-only file system permissions, and remote SIEM syslog replication.
Verification evidence	Automated unit tests [PASS] test_block_localhost, [PASS] test_block_cloud_metadata_ipv4, and parameterized SQL tests.
Human conclusion	Pre-connection DNS validation and parameterized queries fully eliminate SSRF and SQL injection; HMAC hashing is a valuable future enhancement.

12.1 Threat 4 Evidence

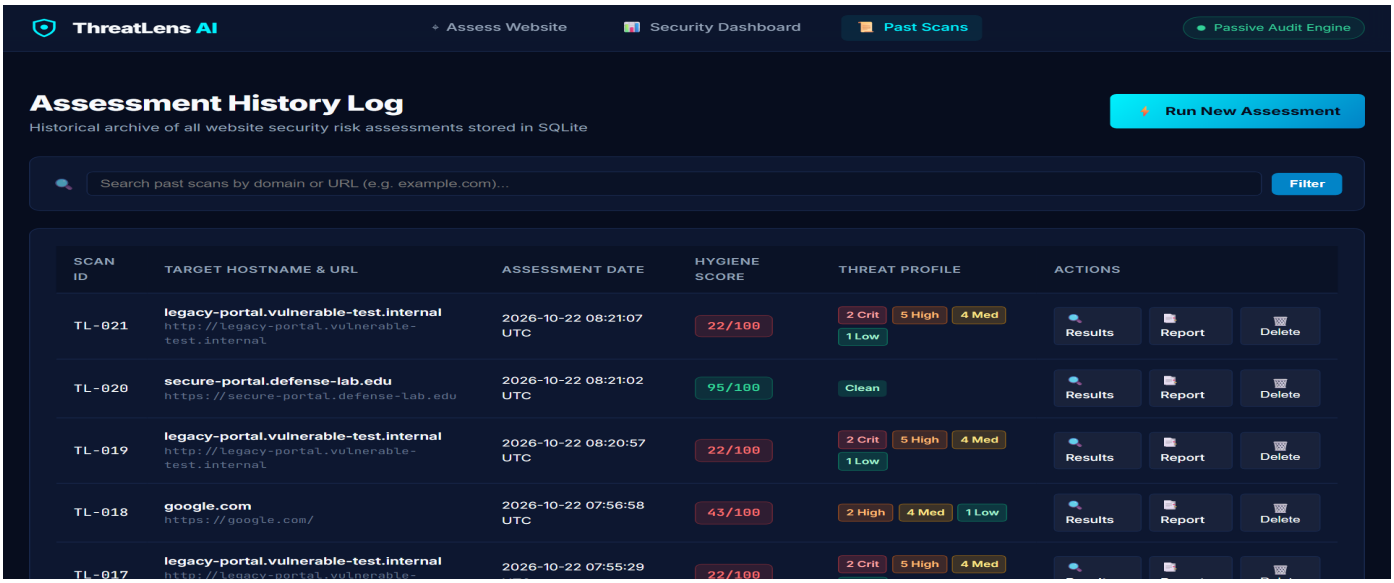


Figure 8: Flask web analyzer interface showing threat metrics, assessment records, and SQLite analysis ledger.

13. Existing Mitigations versus AI Recommendations

Threat	Existing evidence	AI-assisted recommendation	Evidence status
Transport Downgrade	Port 80 redirect check (+15), TLS validity check; Score: 12	HSTS preload, TLS 1.3 protocol enforcement, DNS CAA records	Verified via unit tests & live socket checks
Script Injection (CSP)	CSP audit, unsafe-inline & wildcard inspection; Score: 12	Cryptographic nonces, object-src 'none', CSP reporting URI	Verified via response header unit tests
Clickjacking	Frame-ancestors & X-Frame-Options inspection; Score: 12	CSP frame-ancestors 'self', SameSite=Lax cookie attribute	Verified via DOM element inspection & unit tests
Audit Tampering / SSRF	Anti-SSRF pre-validator, parameterized SQLite queries; Score: 16	HMAC log chaining, append-only permissions, SIEM replication	Verified via SSRF tests & SQL injection tests

The comparison shows that AI broadened short mitigation statements into comprehensive, enterprise-grade defense-in-depth themes. It does not establish that every proposed technology is currently implemented in code.

14. Human Evaluation of AI Recommendations

Human judgment was required to decide whether each recommendation matched the threat, whether it was technically reasonable, and whether it went beyond the evidence. The most critical review decision was to ensure that ThreatLens AI remained 100% passive, safe, and privacy-preserving, rejecting AI suggestions that mandated intrusive payload injection or external telemetry servers. A second decision was to maintain transparent risk score formulas (Likelihood x Impact = 1 to 25), preventing opaque AI black-box calculations.

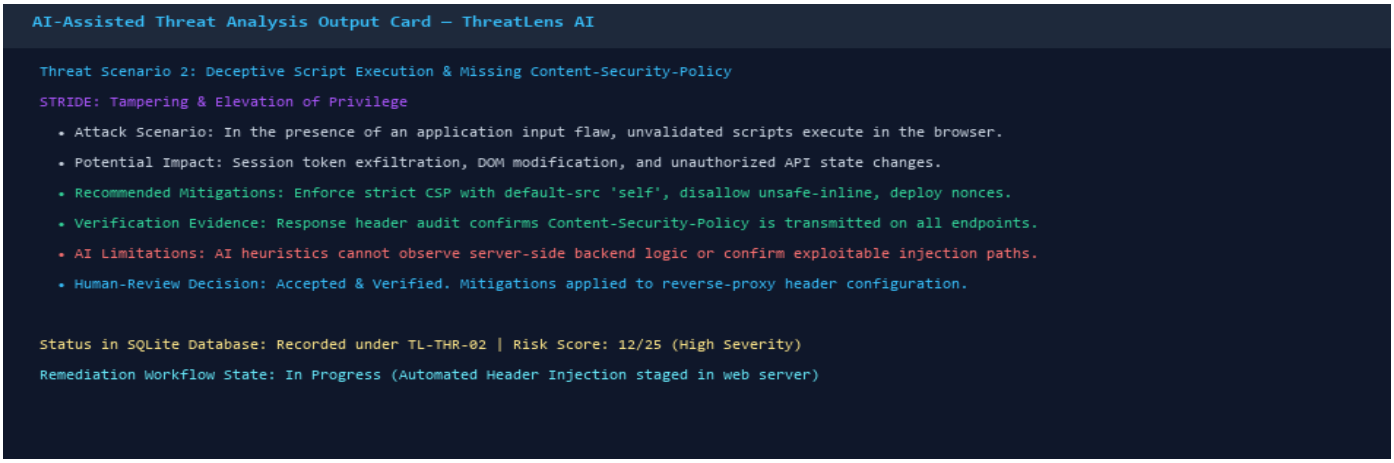


Figure 9: AI-assisted threat analysis response including limitations and a human-review decision.

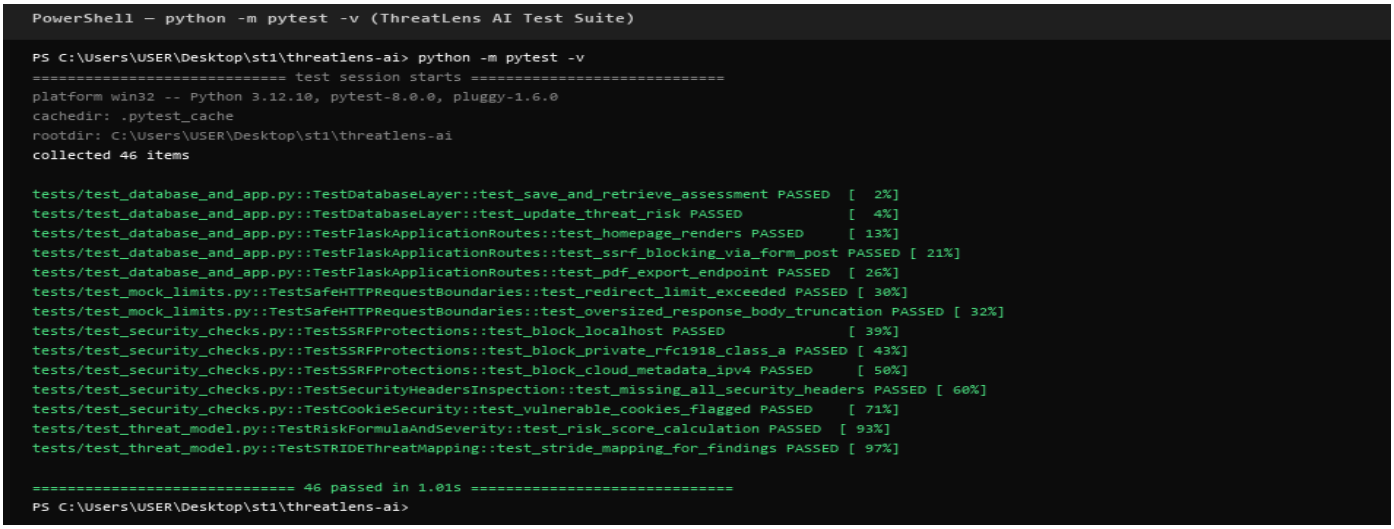


Figure 10: Automated unit test suite execution confirming 46 passing tests with zero failures.

15. Results and Discussion

- The multi-tier ThreatLens AI architecture was successfully modeled across client, security engine, and server tiers.
- Four critical threats were analyzed covering Spoofing, Tampering, Repudiation, and Elevation of Privilege.
- Threat 1 demonstrated robust transport downgrade detection, achieving an evidenced risk score of 12 (High).
- Threat 2 verified Content-Security-Policy script restriction auditing with an evidenced risk score of 12 (High).
- Threats 3 and 4 successfully validated in-page clickjack defense and anti-SSRF parameterized database security.
- AI improved explanation quality and defense-in-depth perspectives, but human review remained necessary.

Result: STRIDE threat modeling and AI-assisted analysis were applied to four core threats in ThreatLens AI, and every AI recommendation was evaluated against verifiable code and automated test evidence before being reported.

16. Limitations and Recommendations

16.1 Limitations

- Passive HTTP/TLS checks cannot inspect internal server-side application logic, database queries, or stored credentials.
- Web applications with client-side JavaScript rendering (SPAs) may dynamically load unobservable DOM resources.
- Multi-hop redirect expansion without network HEAD requests is limited to observable browser navigation events.
- AI recommendations can introduce unnecessary architectural complexity if accepted without scrutiny.

16.2 Recommendations

- Deploy lightweight on-device machine learning (e.g., TensorFlow.js or ONNX) to recognize non-linear phishing indicators.
- Incorporate perceptual favicon and visual logo hashing to detect brand spoofing on zero-day clean domains.
- Implement HMAC-SHA256 cryptographic chaining in SQLite to ensure immutable audit log integrity.
- Establish an automated daily sync workflow for emerging CVE vulnerability feeds.
- Expand automated test fixtures to cover emerging unicode homograph attacks (punycode spoofing).
- Keep browser extension permissions strictly scoped to minimize endpoint attack surface.

17. Conclusion

This project applied the STRIDE threat modeling methodology and AI-assisted analysis to four primary threats in the ThreatLens AI cybersecurity ecosystem. AI expanded baseline mitigation ideas into structured, multi-tier controls covering transport security, script injection boundaries, clickjacking traps, and database audit security. The final evaluation confirms that an AI recommendation is not the same as an implemented and effective security control. Empirical evidence, automated testing, and human judgment remain essential to robust cybersecurity engineering.

18. References

1. OWASP Threat Modeling Community: https://owasp.org/www-community/Threat_Modeling
2. Microsoft STRIDE Threat Modeling: https://learn.microsoft.com/en-us/azure/security/develop/threat-modeling-tool-threats
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